Approximation study: paired differences between models

For each share and parameter budget: the two models' values (mean over seeds) and their difference A - B, for RMSE, maximum absolute error and fit time in seconds. A negative difference in RMSE or max error means model A fits better; a negative seconds difference means A is faster. Pairing: seed s of A with seed s of B (deterministic models paired with every seed); each difference is the mean ± standard deviation of three paired differences. Values in the share's Min-Max price scale.

KAN [1,1] - MLP of t (A = KAN [1,1], B = MLP of t)

share	budget	params A	params B	RMSE A	RMSE B	diff RMSE	max|err| A	max|err| B	diff max|err|	sec A	sec B	diff sec
CBA.AX	15	15	13	0.0707	0.0753	-0.0046 ± 0.0005	0.2201	0.2463	-0.0262 ± 0.0079	1.3	11.8	-10.5 ± 0.5
CBA.AX	30	30	33	0.0464	0.0634	-0.0170 ± 0.0009	0.1804	0.2054	-0.0250 ± 0.0151	0.6	10.5	-10.0 ± 0.1
CBA.AX	60	60	61	0.0272	0.0566	-0.0294 ± 0.0074	0.0954	0.2032	-0.1078 ± 0.0120	0.9	11.7	-10.8 ± 0.6
CBA.AX	120	120	118	0.0172	0.0531	-0.0359 ± 0.0078	0.0804	0.2016	-0.1212 ± 0.0085	1.1	11.0	-9.9 ± 0.6
CBA.AX	240	240	253	0.0131	0.0419	-0.0287 ± 0.0024	0.0696	0.1439	-0.0743 ± 0.0164	2.6	13.3	-10.7 ± 2.6
TSLA	15	15	13	0.0905	0.0868	0.0036 ± 0.0075	0.3084	0.3026	0.0058 ± 0.0052	0.5	11.5	-11.1 ± 0.3
TSLA	30	30	33	0.0495	0.0658	-0.0163 ± 0.0010	0.1920	0.2490	-0.0570 ± 0.0053	0.6	11.7	-11.2 ± 0.2
TSLA	60	60	61	0.0295	0.0614	-0.0318 ± 0.0106	0.1393	0.2456	-0.1062 ± 0.0064	0.7	10.6	-9.8 ± 0.3
TSLA	120	120	118	0.0190	0.0489	-0.0299 ± 0.0028	0.0818	0.2049	-0.1231 ± 0.0302	1.4	11.0	-9.6 ± 0.8
TSLA	240	240	253	0.0138	0.0404	-0.0266 ± 0.0041	0.0721	0.1562	-0.0841 ± 0.0116	1.9	11.5	-9.6 ± 0.2
LLY	15	15	13	0.0491	0.0495	-0.0004 ± 0.0084	0.1440	0.1479	-0.0040 ± 0.0483	0.5	10.5	-10.0 ± 0.1
LLY	30	30	33	0.0261	0.0400	-0.0139 ± 0.0008	0.0790	0.1097	-0.0307 ± 0.0048	1.0	10.5	-9.5 ± 0.7
LLY	60	60	61	0.0173	0.0349	-0.0176 ± 0.0051	0.0635	0.1030	-0.0395 ± 0.0139	0.8	10.8	-10.1 ± 0.4
LLY	120	120	118	0.0120	0.0352	-0.0232 ± 0.0027	0.0483	0.1083	-0.0600 ± 0.0080	1.2	10.9	-9.7 ± 0.1
LLY	240	240	253	0.0083	0.0314	-0.0231 ± 0.0028	0.0342	0.1026	-0.0685 ± 0.0109	1.9	11.4	-9.5 ± 0.7

KAN [1,1] - Fourier MLP (A = KAN [1,1], B = Fourier MLP)

share	budget	params A	params B	RMSE A	RMSE B	diff RMSE	max|err| A	max|err| B	diff max|err|	sec A	sec B	diff sec
CBA.AX	15	15	19	0.0707	0.0842	-0.0136 ± 0.0373	0.2201	0.3608	-0.1407 ± 0.2622	1.3	10.7	-9.3 ± 1.4
CBA.AX	30	30	31	0.0464	0.0579	-0.0115 ± 0.0189	0.1804	0.2290	-0.0486 ± 0.1061	0.6	10.5	-9.9 ± 0.1
CBA.AX	60	60	61	0.0272	0.0258	0.0014 ± 0.0012	0.0954	0.0966	-0.0012 ± 0.0150	0.9	12.0	-11.1 ± 1.7
CBA.AX	120	120	127	0.0172	0.0177	-0.0005 ± 0.0016	0.0804	0.0830	-0.0026 ± 0.0069	1.1	10.7	-9.6 ± 0.5
CBA.AX	240	240	253	0.0131	0.0130	0.0002 ± 0.0014	0.0696	0.0670	0.0025 ± 0.0071	2.6	13.2	-10.6 ± 2.6
TSLA	15	15	19	0.0905	0.0985	-0.0081 ± 0.0398	0.3084	0.3760	-0.0677 ± 0.1913	0.5	11.4	-11.0 ± 0.2

share	budget	params A	params B	RMSE A	RMSE B	diff RMSE	max|err| A	max|err| B	diff max|err|	sec A	sec B	diff sec
TSLA	30	30	31	0.0495	0.0562	-0.0067 ± 0.0114	0.1920	0.2467	-0.0547 ± 0.0614	0.6	12.2	-11.6 ± 1.9
TSLA	60	60	61	0.0295	0.0285	0.0011 ± 0.0022	0.1393	0.1244	0.0149 ± 0.0155	0.7	10.2	-9.5 ± 0.1
TSLA	120	120	127	0.0190	0.0182	0.0008 ± 0.0012	0.0818	0.0812	0.0006 ± 0.0060	1.4	11.3	-10.0 ± 1.8
TSLA	240	240	253	0.0138	0.0143	-0.0005 ± 0.0013	0.0721	0.0676	0.0045 ± 0.0160	1.9	11.0	-9.1 ± 0.4
LLY	15	15	19	0.0491	0.0676	-0.0185 ± 0.0309	0.1440	0.2728	-0.1288 ± 0.1751	0.5	10.4	-9.9 ± 0.2
LLY	30	30	31	0.0261	0.0431	-0.0170 ± 0.0175	0.0790	0.1649	-0.0860 ± 0.0908	1.0	10.9	-9.9 ± 1.5
LLY	60	60	61	0.0173	0.0184	-0.0011 ± 0.0005	0.0635	0.0718	-0.0083 ± 0.0042	0.8	10.5	-9.7 ± 0.2
LLY	120	120	127	0.0120	0.0124	-0.0005 ± 0.0015	0.0483	0.0508	-0.0025 ± 0.0025	1.2	11.2	-10.0 ± 1.2
LLY	240	240	253	0.0083	0.0093	-0.0010 ± 0.0007	0.0342	0.0422	-0.0081 ± 0.0048	1.9	10.8	-8.9 ± 0.7

KAN [1,1] - LSQ spline (A = KAN [1,1], B = LSQ spline)

share	budget	params A	params B	RMSE A	RMSE B	diff RMSE	max|err| A	max|err| B	diff max|err|	sec A	sec B	diff sec
CBA.AX	15	15	15	0.0707	0.0607	0.0100 ± 0.0000	0.2201	0.2258	-0.0057 ± 0.0000	1.3	0.1	1.3 ± 1.5
CBA.AX	30	30	30	0.0464	0.0413	0.0051 ± 0.0000	0.1804	0.1724	0.0080 ± 0.0000	0.6	0.0	0.6 ± 0.1
CBA.AX	60	60	60	0.0272	0.0260	0.0012 ± 0.0000	0.0954	0.1159	-0.0205 ± 0.0000	0.9	0.0	0.9 ± 0.1
CBA.AX	120	120	120	0.0172	0.0173	-0.0002 ± 0.0000	0.0804	0.0824	-0.0020 ± 0.0000	1.1	0.0	1.1 ± 0.2
CBA.AX	240	240	240	0.0131	0.0131	0.0001 ± 0.0000	0.0696	0.0634	0.0061 ± 0.0000	2.6	0.0	2.6 ± 1.6
TSLA	15	15	15	0.0905	0.0661	0.0244 ± 0.0000	0.3084	0.2474	0.0610 ± 0.0000	0.5	0.0	0.5 ± 0.0
TSLA	30	30	30	0.0495	0.0405	0.0090 ± 0.0000	0.1920	0.1665	0.0255 ± 0.0000	0.6	0.0	0.6 ± 0.0
TSLA	60	60	60	0.0295	0.0276	0.0019 ± 0.0000	0.1393	0.1170	0.0223 ± 0.0000	0.7	0.0	0.7 ± 0.1
TSLA	120	120	120	0.0190	0.0189	0.0001 ± 0.0000	0.0818	0.0860	-0.0042 ± 0.0000	1.4	0.1	1.2 ± 0.5
TSLA	240	240	240	0.0138	0.0134	0.0004 ± 0.0000	0.0721	0.0711	0.0010 ± 0.0000	1.9	0.0	1.9 ± 0.0
LLY	15	15	15	0.0491	0.0369	0.0122 ± 0.0000	0.1440	0.1154	0.0286 ± 0.0000	0.5	0.0	0.5 ± 0.0
LLY	30	30	30	0.0261	0.0216	0.0045 ± 0.0000	0.0790	0.0739	0.0051 ± 0.0000	1.0	0.0	1.0 ± 0.6
LLY	60	60	60	0.0173	0.0167	0.0006 ± 0.0000	0.0635	0.0646	-0.0011 ± 0.0000	0.8	0.0	0.8 ± 0.1
LLY	120	120	120	0.0120	0.0114	0.0006 ± 0.0000	0.0483	0.0481	0.0002 ± 0.0000	1.2	0.0	1.2 ± 0.1
LLY	240	240	240	0.0083	0.0081	0.0002 ± 0.0000	0.0342	0.0353	-0.0012 ± 0.0000	1.9	0.0	1.9 ± 0.6

KAN [1,h,1] - KAN [1,1] (A = KAN [1,h,1], B = KAN [1,1])

share	budget	params A	params B	RMSE A	RMSE B	diff RMSE	max|err| A	max|err| B	diff max|err|	sec A	sec B	diff sec
CBA.AX	15	24	15	0.0720	0.0707	0.0013 ± 0.0064	0.2277	0.2201	0.0076 ± 0.0289	1.5	1.3	0.1 ± 1.6
CBA.AX	30	24	30	0.0720	0.0464	0.0256 ± 0.0064	0.2277	0.1804	0.0473 ± 0.0289	1.7	0.6	1.1 ± 0.3
CBA.AX	60	48	60	0.0523	0.0272	0.0251 ± 0.0025	0.1842	0.0954	0.0888 ± 0.0360	5.5	0.9	4.6 ± 0.4
CBA.AX	120	120	120	0.0446	0.0172	0.0274 ± 0.0041	0.1786	0.0804	0.0982 ± 0.0173	6.8	1.1	5.7 ± 0.2
CBA.AX	240	240	240	0.0388	0.0131	0.0257 ± 0.0016	0.1414	0.0696	0.0718 ± 0.0047	9.7	2.6	7.1 ± 1.7
TSLA	15	24	15	0.0877	0.0905	-0.0028 ± 0.0061	0.2912	0.3084	-0.0172 ± 0.0387	1.8	0.5	1.4 ± 0.6
TSLA	30	24	30	0.0877	0.0495	0.0382 ± 0.0061	0.2912	0.1920	0.0993 ± 0.0387	2.3	0.6	1.7 ± 0.8
TSLA	60	48	60	0.0592	0.0295	0.0297 ± 0.0057	0.2247	0.1393	0.0854 ± 0.0154	5.3	0.7	4.6 ± 1.2
TSLA	120	120	120	0.0420	0.0190	0.0230 ± 0.0037	0.1817	0.0818	0.0999 ± 0.0010	7.4	1.4	6.0 ± 0.3
TSLA	240	240	240	0.0407	0.0138	0.0269 ± 0.0044	0.1843	0.0721	0.1122 ± 0.0071	10.4	1.9	8.5 ± 0.2
LLY	15	24	15	0.0481	0.0491	-0.0010 ± 0.0036	0.1472	0.1440	0.0033 ± 0.0178	2.9	0.5	2.4 ± 1.2
LLY	30	24	30	0.0481	0.0261	0.0220 ± 0.0036	0.1472	0.0790	0.0682 ± 0.0178	2.7	1.0	1.8 ± 1.7
LLY	60	48	60	0.0315	0.0173	0.0142 ± 0.0043	0.1001	0.0635	0.0365 ± 0.0170	6.0	0.8	5.3 ± 0.5
LLY	120	120	120	0.0233	0.0120	0.0114 ± 0.0018	0.0809	0.0483	0.0326 ± 0.0084	7.4	1.2	6.2 ± 0.9
LLY	240	240	240	0.0243	0.0083	0.0160 ± 0.0017	0.0774	0.0342	0.0432 ± 0.0073	10.8	1.9	8.8 ± 1.0

Fourier MLP - MLP of t (A = Fourier MLP, B = MLP of t)

share	budget	params A	params B	RMSE A	RMSE B	diff RMSE	max|err| A	max|err| B	diff max|err|	sec A	sec B	diff sec
CBA.AX	15	19	13	0.0842	0.0753	0.0090 ± 0.0377	0.3608	0.2463	0.1145 ± 0.2661	10.7	11.8	-1.2 ± 0.9
CBA.AX	30	31	33	0.0579	0.0634	-0.0055 ± 0.0181	0.2290	0.2054	0.0236 ± 0.0981	10.5	10.5	-0.0 ± 0.1
CBA.AX	60	61	61	0.0258	0.0566	-0.0307 ± 0.0086	0.0966	0.2032	-0.1066 ± 0.0265	12.0	11.7	0.3 ± 2.3
CBA.AX	120	127	118	0.0177	0.0531	-0.0354 ± 0.0088	0.0830	0.2016	-0.1187 ± 0.0142	10.7	11.0	-0.3 ± 0.3
CBA.AX	240	253	253	0.0130	0.0419	-0.0289 ± 0.0022	0.0670	0.1439	-0.0769 ± 0.0093	13.2	13.3	-0.1 ± 0.6
TSLA	15	19	13	0.0985	0.0868	0.0117 ± 0.0472	0.3760	0.3026	0.0734 ± 0.1905	11.4	11.5	-0.1 ± 0.5
TSLA	30	31	33	0.0562	0.0658	-0.0096 ± 0.0122	0.2467	0.2490	-0.0023 ± 0.0662	12.2	11.7	0.4 ± 1.7
TSLA	60	61	61	0.0285	0.0614	-0.0329 ± 0.0094	0.1244	0.2456	-0.1212 ± 0.0106	10.2	10.6	-0.3 ± 0.2
TSLA	120	127	118	0.0182	0.0489	-0.0306 ± 0.0039	0.0812	0.2049	-0.1237 ± 0.0362	11.3	11.0	0.3 ± 1.1
TSLA	240	253	253	0.0143	0.0404	-0.0261 ± 0.0048	0.0676	0.1562	-0.0886 ± 0.0185	11.0	11.5	-0.5 ± 0.2
LLY	15	19	13	0.0676	0.0495	0.0182 ± 0.0390	0.2728	0.1479	0.1248 ± 0.2210	10.4	10.5	-0.1 ± 0.1

share	budget	params A	params B	RMSE A	RMSE B	diff RMSE	max|err| A	max|err| B	diff max|err|	sec A	sec B	diff sec
LLY	30	31	33	0.0431	0.0400	0.0031 ± 0.0182	0.1649	0.1097	0.0552 ± 0.0955	10.9	10.5	0.4 ± 1.1
LLY	60	61	61	0.0184	0.0349	-0.0165 ± 0.0047	0.0718	0.1030	-0.0312 ± 0.0124	10.5	10.8	-0.4 ± 0.2
LLY	120	127	118	0.0124	0.0352	-0.0227 ± 0.0028	0.0508	0.1083	-0.0575 ± 0.0055	11.2	10.9	0.3 ± 1.4
LLY	240	253	253	0.0093	0.0314	-0.0221 ± 0.0025	0.0422	0.1026	-0.0604 ± 0.0090	10.8	11.4	-0.6 ± 0.1

Paired t-tests. Null hypothesis: the mean difference A - B is zero. One-sample t-test on the cell differences (two-sided); 'all 15 cells' pools the three shares and five budgets, per-share rows use that share's five budgets. Columns: number of cells, mean difference, t statistic, p-value.

comparison	metric	scope	n	mean difference	t	p
KAN [1,1] - MLP of t	RMSE	all 15 cells	15	-0.0197	-6.43	< 0.0001
KAN [1,1] - MLP of t	RMSE	CBA.AX	5	-0.0231	-4.17	0.0140
KAN [1,1] - MLP of t	RMSE	TSLA	5	-0.0202	-3.09	0.0365
KAN [1,1] - MLP of t	RMSE	LLY	5	-0.0156	-3.72	0.0205
KAN [1,1] - MLP of t	max |err|	all 15 cells	15	-0.0615	-5.74	< 0.0001
KAN [1,1] - MLP of t	max |err|	CBA.AX	5	-0.0709	-3.55	0.0239
KAN [1,1] - MLP of t	max |err|	TSLA	5	-0.0729	-3.23	0.0320
KAN [1,1] - MLP of t	max |err|	LLY	5	-0.0405	-3.56	0.0236
KAN [1,1] - MLP of t	seconds	all 15 cells	15	-10.12	-68.30	< 0.0001
KAN [1,1] - MLP of t	seconds	CBA.AX	5	-10.36	-54.36	< 0.0001
KAN [1,1] - MLP of t	seconds	TSLA	5	-10.26	-29.14	< 0.0001
KAN [1,1] - MLP of t	seconds	LLY	5	-9.75	-76.13	< 0.0001
KAN [1,1] - Fourier MLP	RMSE	all 15 cells	15	-0.0050	-2.79	0.0145
KAN [1,1] - Fourier MLP	RMSE	CBA.AX	5	-0.0048	-1.51	0.2063
KAN [1,1] - Fourier MLP	RMSE	TSLA	5	-0.0027	-1.38	0.2406
KAN [1,1] - Fourier MLP	RMSE	LLY	5	-0.0076	-1.83	0.1404
KAN [1,1] - Fourier MLP	max |err|	all 15 cells	15	-0.0351	-2.69	0.0174
KAN [1,1] - Fourier MLP	max |err|	CBA.AX	5	-0.0381	-1.40	0.2355
KAN [1,1] - Fourier MLP	max |err|	TSLA	5	-0.0205	-1.21	0.2930
KAN [1,1] - Fourier MLP	max |err|	LLY	5	-0.0467	-1.82	0.1430
KAN [1,1] - Fourier MLP	seconds	all 15 cells	15	-10.01	-50.64	< 0.0001
KAN [1,1] - Fourier MLP	seconds	CBA.AX	5	-10.12	-30.37	< 0.0001
KAN [1,1] - Fourier MLP	seconds	TSLA	5	-10.22	-22.05	< 0.0001
KAN [1,1] - Fourier MLP	seconds	LLY	5	-9.67	-47.50	< 0.0001
KAN [1,1] - LSQ spline	RMSE	all 15 cells	15	0.0047	2.65	0.0189
KAN [1,1] - LSQ spline	RMSE	CBA.AX	5	0.0032	1.67	0.1699

comparison	metric	scope	n	mean difference	t	p
KAN [1,1] - LSQ spline	RMSE	TSLA	5	0.0072	1.56	0.1947
KAN [1,1] - LSQ spline	RMSE	LLY	5	0.0036	1.57	0.1905
KAN [1,1] - LSQ spline	max |err|	all 15 cells	15	0.0082	1.64	0.1235
KAN [1,1] - LSQ spline	max |err|	CBA.AX	5	-0.0028	-0.55	0.6121
KAN [1,1] - LSQ spline	max |err|	TSLA	5	0.0211	1.83	0.1406
KAN [1,1] - LSQ spline	max |err|	LLY	5	0.0063	1.11	0.3280
KAN [1,1] - LSQ spline	seconds	all 15 cells	15	1.12	7.08	< 0.0001
KAN [1,1] - LSQ spline	seconds	CBA.AX	5	1.29	3.73	0.0202
KAN [1,1] - LSQ spline	seconds	TSLA	5	0.99	3.69	0.0210
KAN [1,1] - LSQ spline	seconds	LLY	5	1.08	4.42	0.0115
KAN [1,h,1] - KAN [1,1]	RMSE	all 15 cells	15	0.0188	6.04	< 0.0001
KAN [1,h,1] - KAN [1,1]	RMSE	CBA.AX	5	0.0210	4.25	0.0131
KAN [1,h,1] - KAN [1,1]	RMSE	TSLA	5	0.0230	3.33	0.0292
KAN [1,h,1] - KAN [1,1]	RMSE	LLY	5	0.0125	3.29	0.0302
KAN [1,h,1] - KAN [1,1]	max |err|	all 15 cells	15	0.0585	5.63	< 0.0001
KAN [1,h,1] - KAN [1,1]	max |err|	CBA.AX	5	0.0627	3.85	0.0182
KAN [1,h,1] - KAN [1,1]	max |err|	TSLA	5	0.0759	3.21	0.0326
KAN [1,h,1] - KAN [1,1]	max |err|	LLY	5	0.0368	3.53	0.0242
KAN [1,h,1] - KAN [1,1]	seconds	all 15 cells	15	4.36	6.05	< 0.0001
KAN [1,h,1] - KAN [1,1]	seconds	CBA.AX	5	3.74	2.79	0.0493
KAN [1,h,1] - KAN [1,1]	seconds	TSLA	5	4.44	3.31	0.0297
KAN [1,h,1] - KAN [1,1]	seconds	LLY	5	4.89	3.78	0.0194
Fourier MLP - MLP of t	RMSE	all 15 cells	15	-0.0146	-3.16	0.0069
Fourier MLP - MLP of t	RMSE	CBA.AX	5	-0.0183	-2.14	0.0992
Fourier MLP - MLP of t	RMSE	TSLA	5	-0.0175	-2.09	0.1044
Fourier MLP - MLP of t	RMSE	LLY	5	-0.0080	-1.00	0.3753
Fourier MLP - MLP of t	max |err|	all 15 cells	15	-0.0264	-1.18	0.2567
Fourier MLP - MLP of t	max |err|	CBA.AX	5	-0.0328	-0.74	0.5019

comparison	metric	scope	n	mean difference	t	p
Fourier MLP - MLP of t	max |err|	TSLA	5	-0.0525	-1.37	0.2434
Fourier MLP - MLP of t	max |err|	LLY	5	0.0062	0.17	0.8726
Fourier MLP - MLP of t	seconds	all 15 cells	15	-0.12	-1.01	0.3303
Fourier MLP - MLP of t	seconds	CBA.AX	5	-0.24	-0.97	0.3865
Fourier MLP - MLP of t	seconds	TSLA	5	-0.03	-0.18	0.8629
Fourier MLP - MLP of t	seconds	LLY	5	-0.07	-0.38	0.7251

Outcome in brief

• KAN [1,1] vs plain MLP: the KAN fits better. Mean RMSE difference -0.0197 ($t = -6.43$, $p < 0.0001$); negative in 14 of 15 cells.
• KAN [1,1] vs Fourier MLP: pooled over all budgets the KAN is slightly better (mean -0.0050, $t = -2.79$, $p = 0.0145$), but the difference comes from the 15- and 30-parameter cells; from 60 parameters upward the two tie (mean 0.0000, $t = -0.05$, $p = 0.96$).
• KAN [1,1] vs least-squares spline: the spline is marginally better pooled (mean 0.0047, $t = 2.65$, $p = 0.0189$), again only at the two smallest budgets; from 60 upward the two agree (mean 0.0005, $p = 0.04$).
• KAN [1,h,1] vs KAN [1,1]: the hidden layer is worse. Mean RMSE difference 0.0188 ($t = 6.04$, $p < 0.0001$); positive in every cell.
• Fourier MLP vs plain MLP: the Fourier features help. Mean RMSE difference -0.0146 ($t = -3.16$, $p = 0.0069$).
• Seconds: the KAN [1,1] fits are about 10 s faster than either MLP ($p < 0.0001$) and about 1.1 s slower than the direct spline solve; the hidden-layer KAN is about 4.4 s slower than the one-edge KAN.

Figure. Every cell difference (coloured points, one per share and budget) and the pooled mean with its 95% confidence interval (black diamond and bar), for each comparison and metric. Points below the dotted zero line favour model A.

